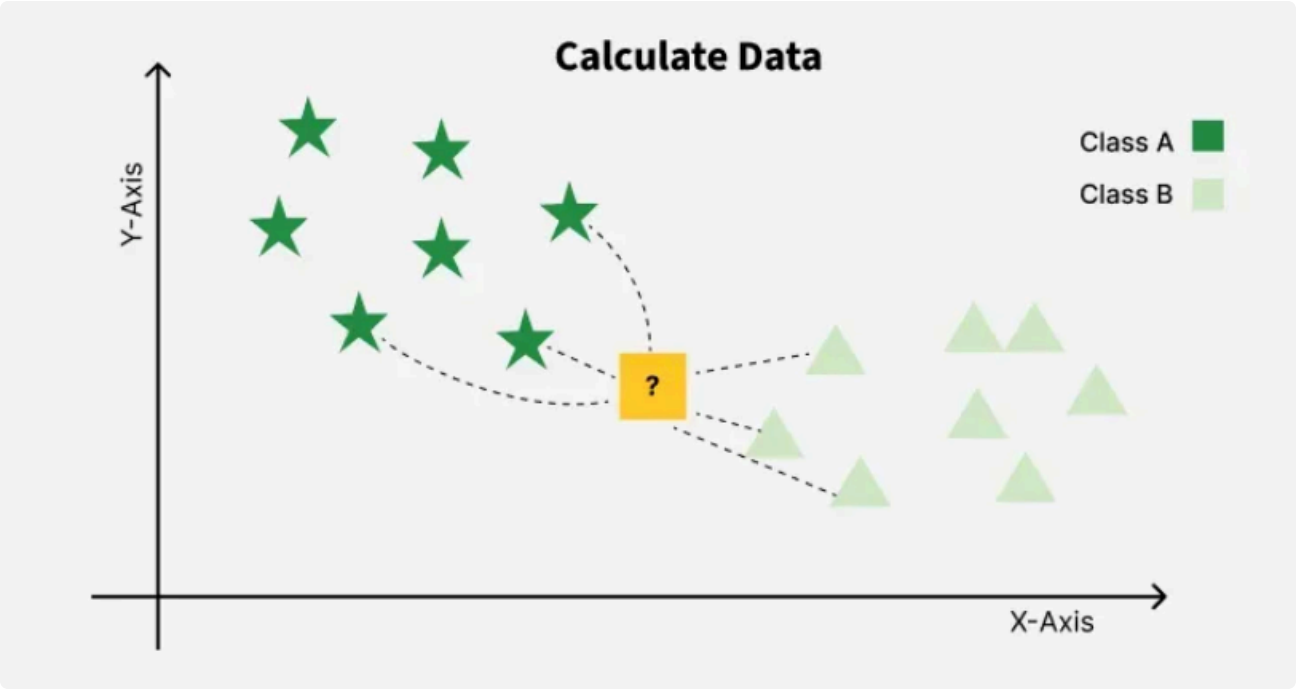
What it is

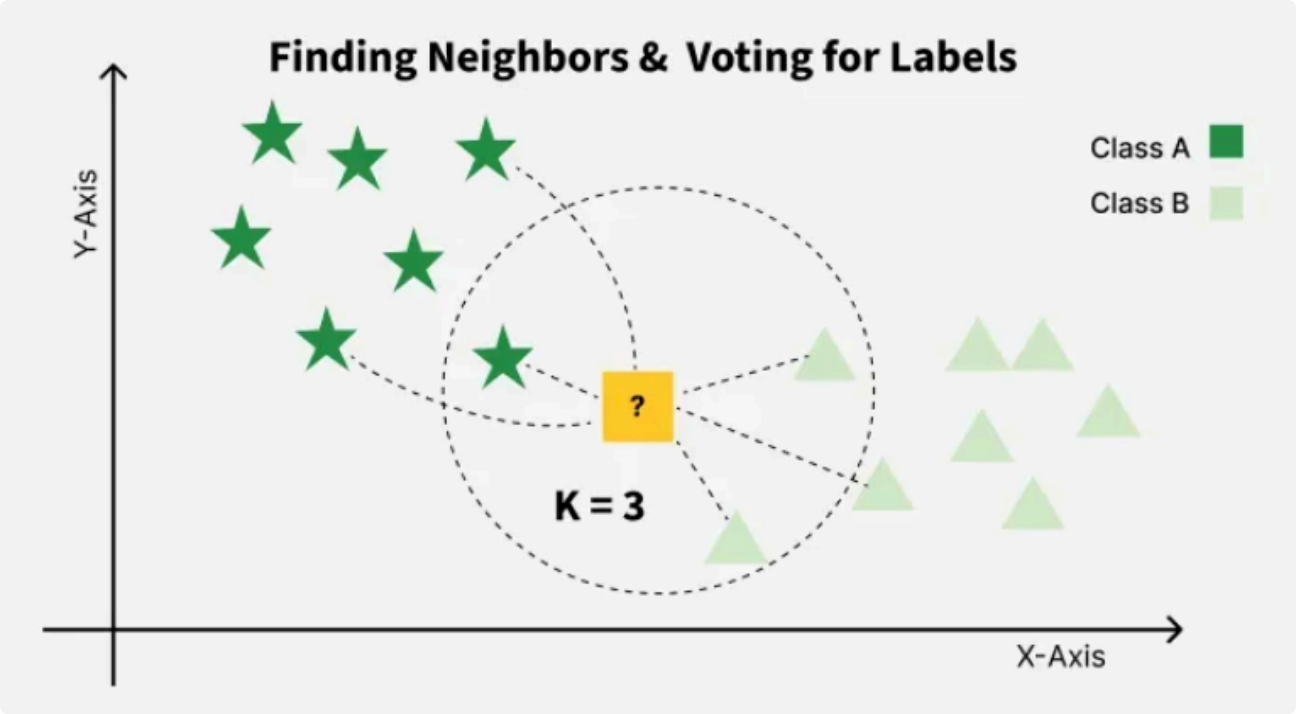
KNN predicts the answer for a new data point by looking at the most similar data points it has already seen, and copying their answer.

It can do two kinds of prediction:

- **Classification** (predicting a category, such as Bought / Not bought): take the majority vote of the similar points.
- **Regression** (predicting a number, such as a house price): take the average of the similar points.







Note :
KNN is also called as a lazy learner algorithm because it does not learn from the training set immediately instead it stores the entire dataset and performs computations only at the time of classification.

What “K” and “nearest neighbors” mean

- **Neighbors** are existing data points whose answers are already known.
- **Nearest** means most similar to the new point. Similarity is measured as distance: the smaller the distance between two points’ values, the more similar they are.

- **K** is how many nearest points you look at. K = 3 means “look at the 3 most similar points.”

****Example:**** Imagine you're deciding which fruit it is based on its shape and size. You compare it to fruits you already know.

- If k = 3, the algorithm looks at the 3 closest fruits to the new one.
- If 2 of those 3 fruits are apples and 1 is a banana, the algorithm says the new fruit is an apple because most of its neighbors are apples.

How to choose the value of k for KNN Algorithm?

- The value of k in KNN decides how many neighbors the algorithm looks at when making a prediction.
- Choosing the right k is important for good results.
- If the data has lots of noise or outliers, using a larger k can make the predictions more stable.
- But if k is too large the model may become too simple and miss important patterns and this is called underfitting.
- So k should be picked carefully based on the data.

****Statistical Methods for Selecting k****

- ****Cross-Validation****: Cross-Validation is a good way to find the best value of k is by using k-fold cross-validation. This means dividing the dataset into k parts. The model is trained on some of these parts and tested on the remaining ones. This process is repeated for each part.
- ****Elbow Method****: In Elbow Method we draw a graph showing the error rate or accuracy for different k values. As k increases the error usually drops at first. But after a certain point error stops decreasing quickly. The point where the curve changes direction and looks like an "elbow" is usually the best choice for k.
- ****Odd Values for k****: It’s a good idea to use an odd number for k especially in classification problems. This helps avoid ties when deciding which class is the most common among the neighbors.

How a prediction is made

1. A new data point arrives with no answer.
2. Calculate the distance from the new point to every existing point.
3. Sort the existing points from smallest distance to largest.
4. Pick the top K.

5. Take the majority vote (classification) or the average (regression). That is the prediction.

Where KNN is not used

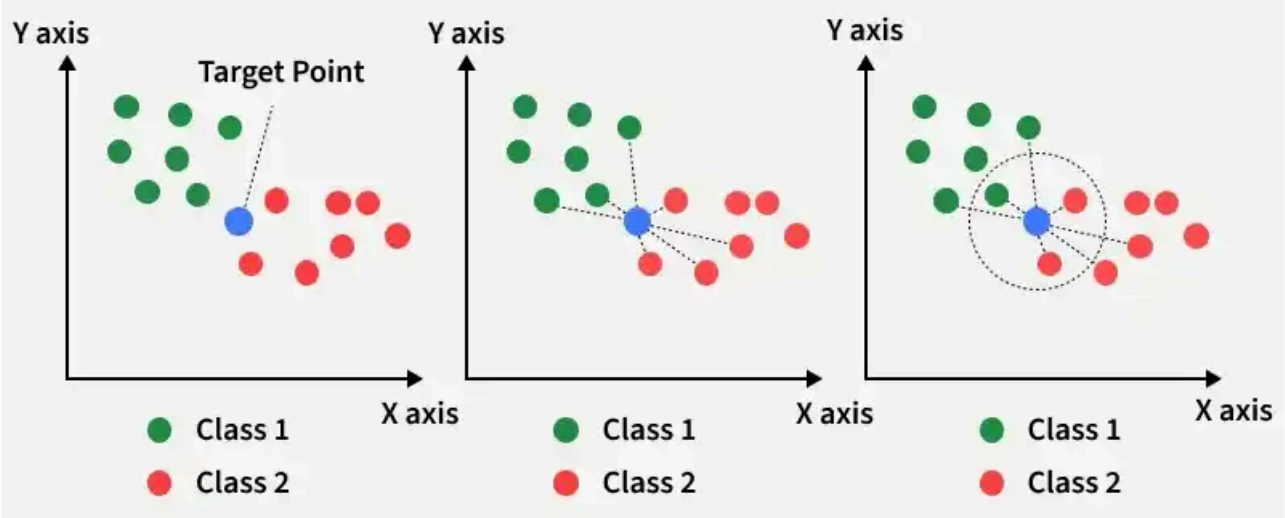
- **Very large datasets:** every prediction compares the new point against all stored points, so it gets slow.
- **Data with many columns:** with hundreds of columns, all points end up roughly equally far apart, so “nearest” stops meaning much.
- **Data with many irrelevant columns or noisy labels:** they distort the distances and the votes.
- **Heavily imbalanced data:** if 95% of points are one category, that category dominates the vote.
- **Systems needing instant predictions** at high volume.

Advantages

- Very easy to understand and explain
- No real training step: it just stores the data
- Works for both classification and regression
- Only one main setting to tune (K)
- New data can be added without rebuilding anything

Disadvantages

- Slow at prediction time on large data
- Needs all the data kept in memory
- Requires scaling
- Sensitive to irrelevant columns and unusual points
- Does not tell you which columns matter or why



Step 1: Selecting the optimal value of K

- K is the number of nearest neighbors considered for prediction.

Step 2: Calculating distance

- To measure the similarity between target and training data points Euclidean distance is widely used. Distance is calculated between data points in the dataset and target point.

Step 3: Finding Nearest Neighbors

- The k data points with the smallest distances to the target point are nearest neighbors.

Step 4: Voting for Classification or Taking Average for Regression

- When you want to classify a data point into a category like spam or not spam, the KNN algorithm looks at the K closest points in the dataset. These closest points are called neighbors. The algorithm then looks at which category the neighbors belong to and picks the one that appears the most. This is called majority voting.
- In regression, the algorithm still looks for the K closest points. But instead of voting for a class in classification, it takes the average of the values of those K neighbors. This average is the predicted value for the new point for the algorithm.